AI-Based Supply Chain Automation

Introduction

This project focuses on supply chain systems, which generate large volumes of data from orders, inventory, shipments, and suppliers. However, much of this data remains underutilized due to manual processes and lack of automation.

The project leverages modern AI-powered tools—Quadratic and n8n—to build an end-to-end data analysis pipeline for supply chain optimization.

By integrating automation with intelligent data analysis, the project demonstrates how raw operational data can be transformed into meaningful insights. It presents a practical approach to combining data engineering, analytics, and business understanding to improve efficiency and decision-making in supply chain management.

Problem Statement

Supply chain systems often struggle with inefficient data management and lack of automation. Large volumes of data generated from orders, inventory, shipments, and suppliers are frequently handled manually, leading to errors, delays, and inconsistencies. Additionally, data is often scattered across multiple systems, making it difficult to gain a unified and real-time view of operations.

These challenges result in issues such as poor demand forecasting, overstocking or stockouts, delayed deliveries, and increased operational costs.

The core problem addressed in this project is:

How to automate the collection, processing, and analysis of supply chain data to generate accurate, real-time insights that support better decision-making and improve overall efficiency.

Methodology

Step 1:Connecting n8n to Gmail (Filter by Specific Subject)

The first step in the project involves setting up an automated workflow using n8n and Gmail to collect relevant supply chain data from emails based on a specific subject line.

To automatically fetch emails that contain important supply chain information (e.g., orders, shipment updates) by filtering them using a predefined subject.

1. Create a New Workflow in n8n
    - Open n8n and create a new workflow.
    - Add a trigger node to start the automation.
2. Add Gmail Trigger Node
    - Select the Gmail node in n8n.
    - Choose the trigger type (e.g., *New Email*).
3. Authenticate Gmail Account
    - Connect your Gmail account securely using OAuth credentials.
    - Grant necessary permissions to allow n8n to read emails.
4. Set Email Filter (Subject-Based)
    - Configure the node to filter emails by a specific subject (e.g., “Order Update” or “Shipment Details”).
    - This ensures only relevant emails are processed.
5. Define Polling or Trigger Frequency

- Set how frequently n8n should check for new emails (e.g., every minute or hourly).

6. Test the Workflow
   - Send a test email with the defined subject.

Step 2: Connecting Supabase with PostgreSQL

The second step focuses on setting up a centralized database to store and manage the extracted data. This is achieved by connecting Supabase with PostgreSQL.

To store structured supply chain data in a reliable database for further analysis and processing.

1. Create a Project in Supabase
   - Sign in to Supabase and create a new project.
   - This automatically provisions a PostgreSQL database.
2. Access Database Credentials
   - Go to project settings in Supabase.
   - Copy the required details:
     - Host
     - Database name
     - Username
     - Password
     - Port
3. Understand Connection Type
   - Supabase provides two types:
     - Direct database connection
     - Connection via pooler (PgBouncer)
   - Choose the appropriate one depending on your use case (n8n usually works well with pooled connections).
4. Set Up Database Tables
   - Create tables to store incoming data (e.g., orders, shipments).
   - Define columns like:
     - Order ID
     - Product Name
     - Quantity
     - Status
     - Timestamp
5. Connect Supabase to n8n
   - In n8n, add a PostgreSQL node.
   - Enter Supabase database credentials.
   - Test the connection to ensure it works properly.
6. Insert Data into Database
   - Configure the node to insert incoming email data into the tables.
   - Map fields from Gmail data to database columns.
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Step 4: Methodology – AI-Driven KPI Analysis in Quadratic

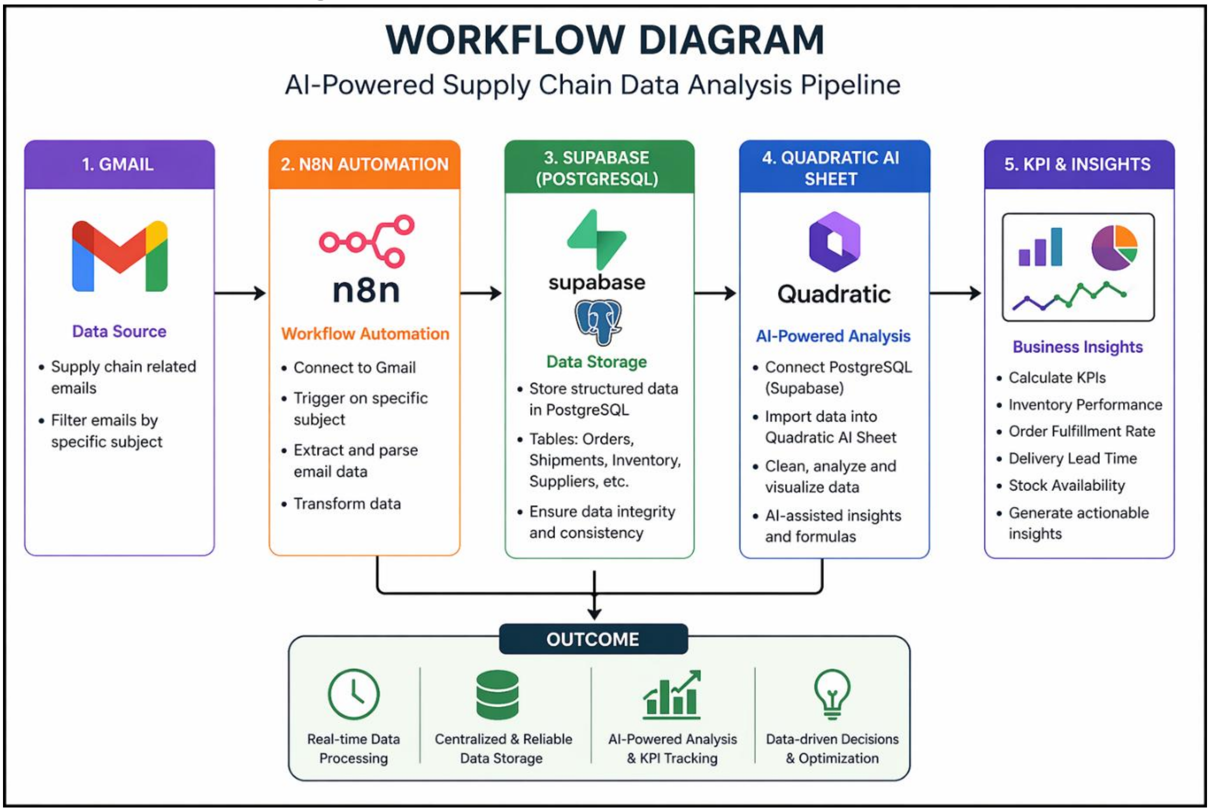
The fourth step focuses on using AI capabilities within Quadratic to calculate and analyze key performance indicators (KPIs) for the supply chain.

To evaluate supply chain performance by generating and analyzing important KPIs using AI-powered data analysis.

1. Identify Key KPIs
   - Define the most relevant metrics for supply chain performance, such as:
     - Inventory Turnover

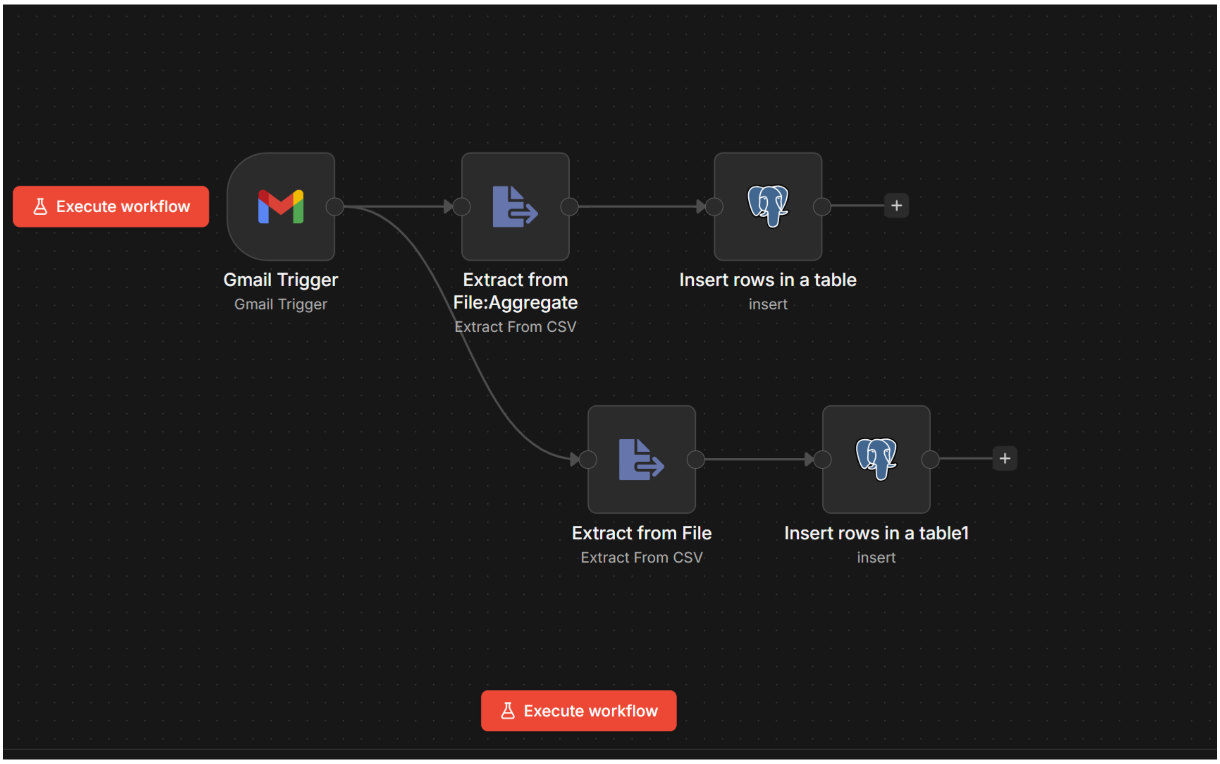
        - Order Fulfillment Rate
        - Delivery Time (Lead Time)
        - Stock Availability
2. Load Data into Quadratic
    - Use the connected PostgreSQL (Supabase) data.
    - Ensure all required fields are available for KPI calculations.
3. Apply AI-Assisted Calculations
    - Use formulas or Python-like queries in Quadratic to compute KPIs.
    - Leverage AI suggestions for faster and more accurate calculations.
4. Analyze Trends and Patterns
    - Observe KPI changes over time.
    - Identify patterns such as delays, demand spikes, or inefficiencies.
5. Generate Visual Insights
    - Create charts and graphs to represent KPIs clearly.
    - Example:
        - Line chart for delivery time trends
        - Bar chart for inventory levels
6. Interpret Results
    - Identify problem areas (e.g., frequent delays, low stock levels).
    - Highlight opportunities for optimization.

Workflow Diagram

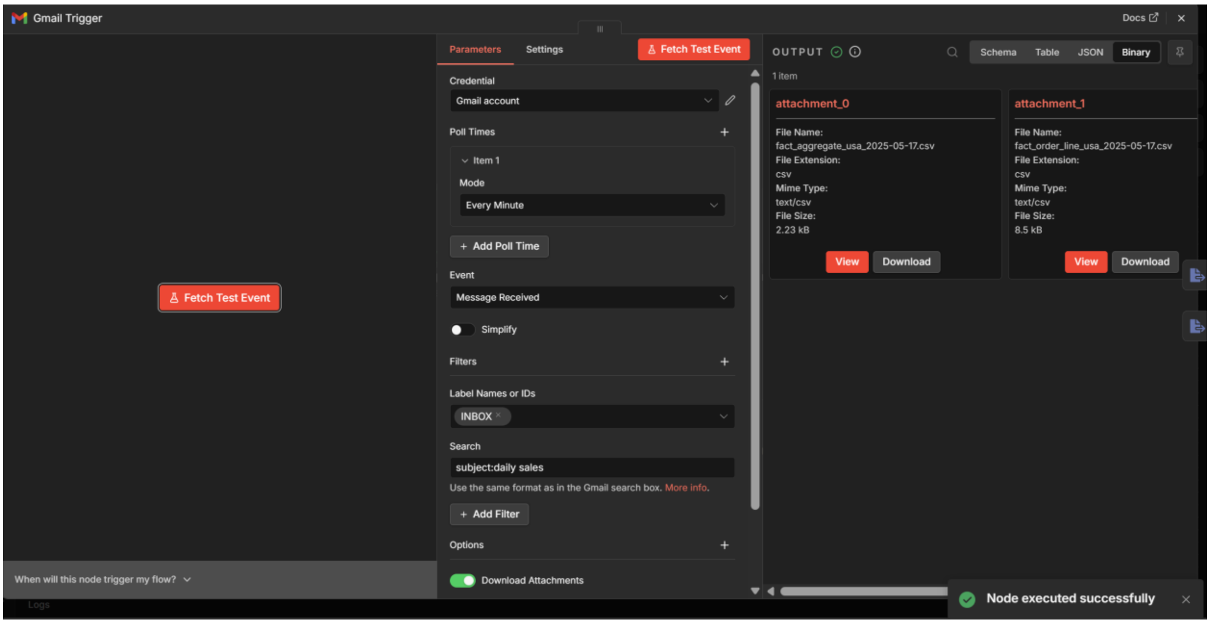


Screenshots

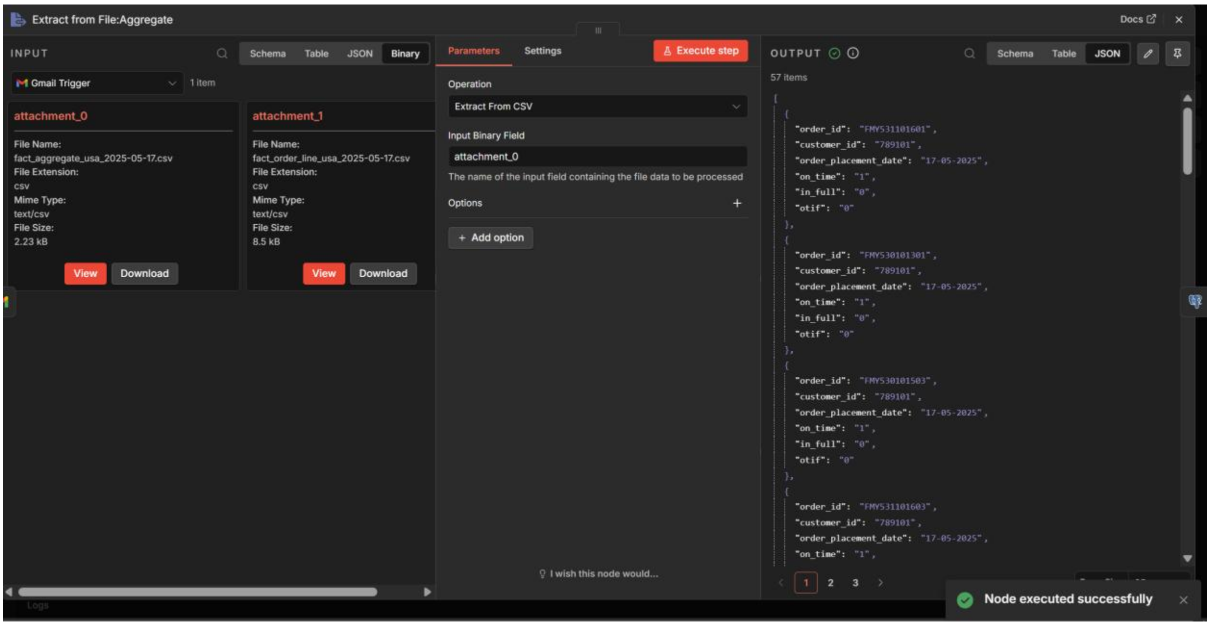
1.n8n-Based Email-to-Database Data Pipeline Workflow1



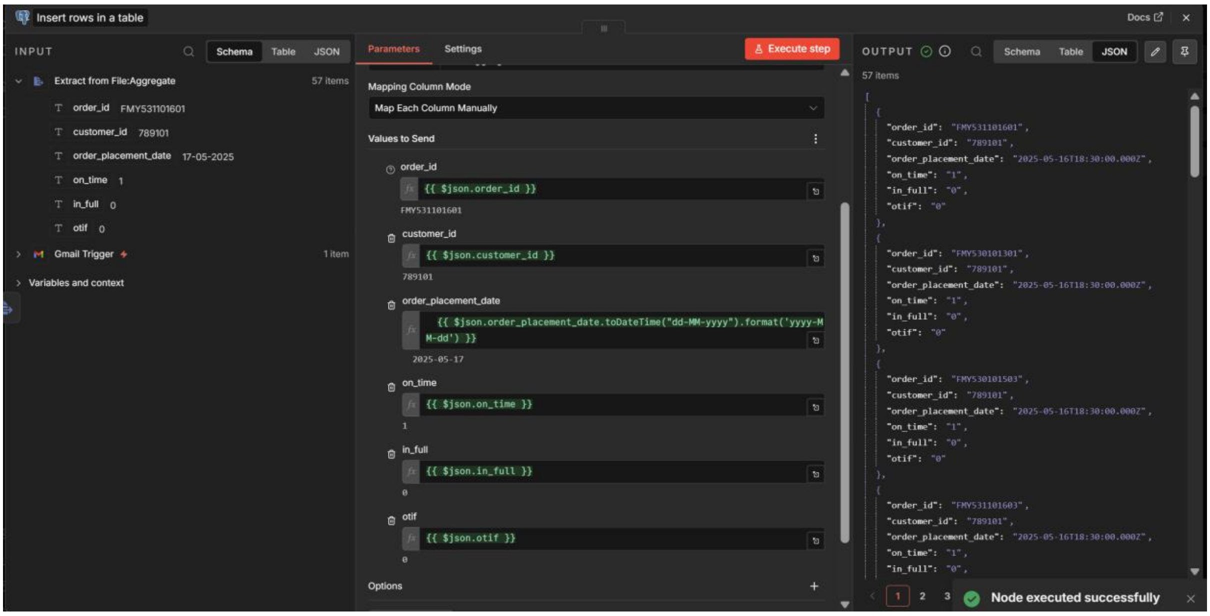
2.Gmail Trigger Setup in n8n for Fetching Email Attachments



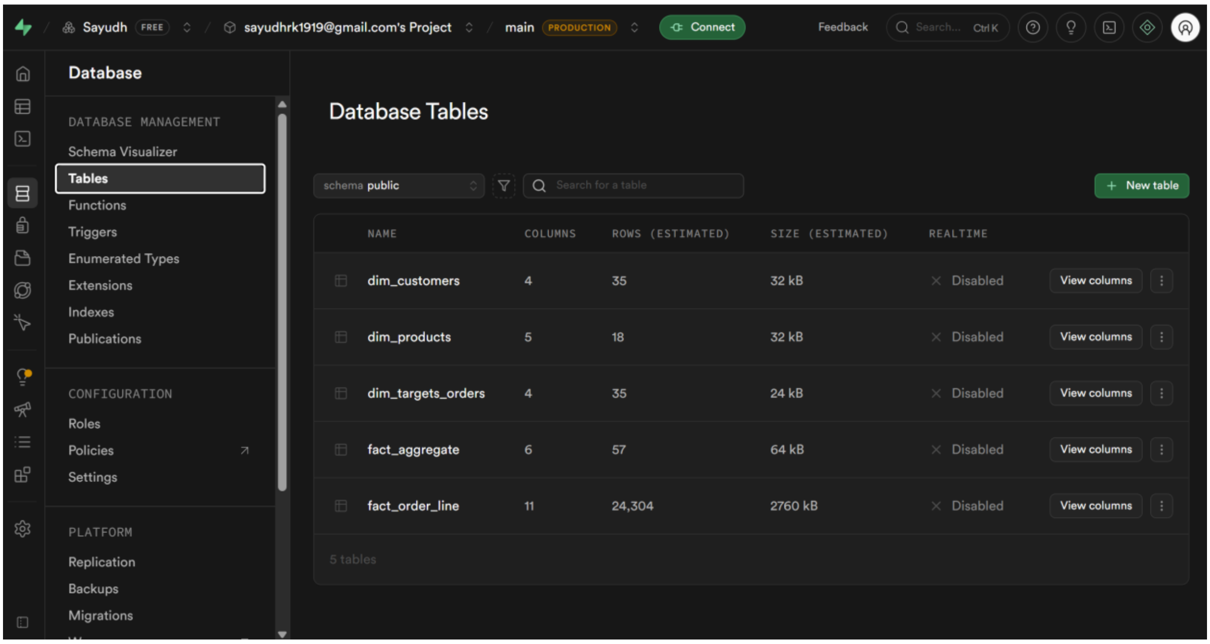
3.CSV Data Extraction in n8n using Extract from File Node



4.Data Insertion into PostgreSQL Table using n8n with Field Mapping



5.Database Tables Overview in Supabase PostgreSQL



6.Data Analysis in Quadratic AI Sheet Connected to PostgreSQL